

4.7.3 Induced potential, transformers and the National Grid (physics only) (HT only)

4.7.3.1 Induced potential (HT only)

Content	Key opportunities for skills development
If an electrical conductor moves relative to a magnetic field or if there is a change in the magnetic field around a conductor, a potential difference is induced across the ends of the conductor. If the conductor is part of a complete circuit, a current is induced in the conductor. This is called the generator effect. An induced current generates a magnetic field that opposes the original change, either the movement of the conductor or the change in magnetic field. Students should be able to recall the factors that affect the size of the induced potential difference/induced current. Students should be able to recall the factors that affect the direction of the induced potential difference/induced current. Students should be able to apply the principles of the generator effect in a given context.	

4.7.3.2 Uses of the generator effect (HT only)

Content	Key opportunities for skills development
The generator effect is used in an alternator to generate ac and in a dynamo to generate dc. Students should be able to:	
- explain how the generator effect is used in an alternator to generate ac and in a dynamo to generate dc - draw/interpret graphs of potential difference generated in the coil against time.	WS 1.4

4.7.3.3 Microphones (HT only)

Content	Key opportunities for skills development
Microphones use the generator effect to convert the pressure variations in sound waves into variations in current in electrical circuits. Students should be able to explain how a moving-coil microphone works.	

4.7.3.4 Transformers (HT only)

Content	Key opportunities for skills development
A basic transformer consists of a primary coil and a secondary coil wound on an iron core. Iron is used as it is easily magnetised. Knowledge of laminations and eddy currents in the core is not required. The ratio of the potential differences across the primary and secondary coils of a transformer V_p and V_s depends on the ratio of the number of turns on each coil, n_p and n_s.	
$\left[\frac{V_p}{V_s} = \frac{n_p}{n_s} \right]$ potential difference, V_p and V_s in volts, V In a step-up transformer $V_s > V_p$ In a step-down transformer $V_s < V_p$ If transformers were 100% efficient, the electrical power output would equal the electrical power input.	MS 3b, c Students should be able to apply this equation which is given on the Physics equation sheet.
$V_s \times I_s = V_p \times I_p$ Where $V_s \times I_s$ is the power output (secondary coil) and $V_p \times I_p$ is the power input (primary coil). power input and output, in watts, W Students should be able to: • explain how the effect of an alternating current in one coil in inducing a current in another is used in transformers • explain how the ratio of the potential differences across the two coils depends on the ratio of the number of turns on each • calculate the current drawn from the input supply to provide a particular power output	MS 3b, c Students should be able to apply this equation which is given on the Physics equation sheet.
• apply the equation linking the p.d.s and number of turns in the two coils of a transformer to the currents and the power transfer involved, and relate these to the advantages of power transmission at high potential differences.	MS 1c, 3b, c